

**SELF-MONITORING AND ANONYMOUS
COUNSELLING BASED DIGITAL COMPANION FOR
ACADEMIC STRESS MANAGEMENT**

25-26J-480

Project Proposal Report

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Sri Lanka Institute of Information Technology Sri Lanka

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**EMOTIONALLY INTELLIGENT PERSONAL
ASSISTANT FOR REAL-TIME STRESS DETECTION
AND ADAPTIVE THERAPEUTIC SUPPORT**

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Academic stress is a growing concern among university students, often leading to anxiety, depression, emotional exhaustion, and reduced academic performance. Traditional counseling services are constrained by limited availability, social stigma, and scalability challenges. To address this gap, this research proposes Self-Monitoring and Anonymous Counselling Based Digital Companion for Academic Stress Management.

The system introduces an AI-powered chatbot as the central digital companion, enabling students to anonymously express their feelings through text-based interactions. It integrates a mobile-based input interface where users provide text, mood, and emotional feedback. The processing pipeline includes data preparation techniques such as tokenization, cleaning, and normalization, followed by advanced emotion and stress detection models using Natural Language Processing (NLP) and deep learning approaches. Once the emotional state is identified, the chatbot generates adaptive therapeutic conversations, assigns stress-relief tasks, and verifies task engagement. In cases where elevated risk signals such as persistent hopelessness or self-harm intent are detected, the system escalates the case to human counselors while preserving anonymity.

The expected outcome is a scalable, accessible, and empathetic digital support tool capable of detecting emotional distress and providing evidence-based coping strategies. Anticipated results include improved emotional regulation, reduced stress levels, and higher engagement compared to traditional self-help tools. Furthermore, the system is expected to enhance task adherence through adaptive monitoring and provide a proactive safety net for students at risk. This research highlights the potential of combining affective computing, personalized task assignment, and escalation workflows into one integrated platform. It is anticipated to contribute both to academic knowledge in human-centered AI systems and to practical applications in student mental health support. Recommendations will focus on ensuring ethical safeguards, cultural sensitivity, and scalability for deployment across diverse educational settings.

Keywords: Emotion Detection, Stress Monitoring, Therapeutic Chatbot, Adaptive Interventions, Student Mental Health

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CBT	Cognitive Behavioural Therapy
NLP	Natural Language Processing
HCI	Human-Computer Interaction

1. INTRODUCTION

The research project, Self-Monitoring and Anonymous Counselling Based Digital Companion for Academic Stress Management, aims to provide university students with an accessible, scalable, and anonymous tool to manage academic stress. Academic stress is a pervasive and growing concern among university students, often manifesting as anxiety, depression, burnout, and reduced academic performance. Surveys indicate that over 75% of students report experiencing significant stress, with many facing associated issues such as insomnia, panic attacks, and emotional exhaustion [1]. Traditional counselling services, though beneficial, face limitations including restricted availability, social stigma, and challenges in scaling to meet the needs of large student populations [2]. Consequently, there is an urgent need for innovative digital solutions to support students' mental health effectively.

In recent years, artificial intelligence (AI)-powered chatbots have emerged as promising tools for mental health support. Conversational agents such as Woebot, based on Cognitive Behavioural Therapy (CBT), have demonstrated measurable reductions in anxiety and depression symptoms within weeks of interaction [3]. Tess, another psychological AI, has shown improvements in students' emotional well-being through supportive conversations [4]. Similarly, platforms like Wysa and EMMA combine empathetic dialogue with emotion-aware responses to increase engagement and adherence to therapeutic interventions [5], [6]. These studies show that emotionally intelligent chatbots can play a vital role in managing academic stress.

Developing effective digital companions requires mastery of several key areas. Natural Language Processing (NLP) techniques, including transformer-based models such as BERT and RoBERTa, enable accurate detection of emotional states and stress patterns from short text inputs [7]. Affective computing provides theoretical frameworks for interpreting mood, arousal, and sentiment from user interactions [8]. On the engineering side, mobile app development using frameworks like React Native and cloud-based backend services such as Firebase ensure cross-platform scalability, security, and real-time performance [9]. Furthermore, principles of human-computer interaction (HCI) and user-centered design are essential to make the chatbot engaging, accessible, and culturally sensitive for diverse student populations [10].

The current state of the art in digital mental health companions combines machine learning with evidence-based therapeutic frameworks. Systems like Wysa incorporate coping exercises and empathetic dialogue, while EMMA adjusts tone and pacing based on the user's detected emotions [5], [6]. Despite these advancements, existing solutions often overlook two critical areas: (1) assigning personalized coping tasks tailored to the student's needs, and (2) verifying whether students complete the suggested stress-relief tasks. As a result, interventions may remain generic and less effective in addressing the real-life experiences of students.

This research proposes a Self-Monitoring and Anonymous Counselling Based Digital Companion for Academic Stress Management. The system introduces an AI-powered chatbot capable of real-time stress and emotion detection and adaptive therapeutic conversations. Its key features include personalized stress-relief task assignment with monitoring and verification, ensuring that interventions are actively completed, and a structured escalation mechanism that alerts human counsellors in cases of persistent hopelessness or suicidal ideation, while preserving student anonymity.

By integrating these features, the proposed system advances beyond prior work by delivering a holistic, adaptive, and ethically grounded solution tailored for university students. It combines state-of-the-art NLP and affective computing techniques with practical therapeutic frameworks, providing both scientific rigor and real-world applicability. The expected outcomes include enhanced emotional regulation, reduced stress levels, improved task adherence, and a proactive safety net for students at risk. Furthermore, this research contributes to the academic literature on human-centered AI systems and offers a scalable solution for student mental health support across educational settings.

2. BACKGROUND AND LITERATURE SURVEY

Background

Academic stress has emerged as one of the most pressing concerns in higher education, with university students reporting significantly higher levels of psychological distress compared to the general population. According to the American College Health Association, more than 75% of students experience substantial levels of stress during their academic careers, with many also suffering from associated conditions such as insomnia, panic attacks, depression, and emotional exhaustion [1]. These conditions negatively impact not only academic performance but also long-term mental health and overall quality of life.

Stress among students is often triggered by academic workloads, examination pressure, competition, and future career concerns [11]. While counseling services are available in many universities, their effectiveness is constrained by three key challenges: (1) limited availability of qualified professionals, (2) persistent social stigma attached to seeking psychological help, and (3) difficulties in scaling these services to support large student populations [2]. As a result, many students either do not seek help at all or fail to receive timely intervention.

In parallel, the rise of digital health solutions has opened new opportunities for mental health support. With the widespread adoption of mobile devices and internet-based platforms, students are increasingly turning to online tools and self-help resources. Among these, Artificial Intelligence (AI)-powered chatbots have attracted particular attention due to their potential for anonymity, scalability, and 24/7 availability. These systems allow students to engage in private, stigma-free conversations while receiving evidence-based therapeutic support.

The growing field of affective computing further strengthens this potential. By enabling machines to recognize, interpret, and respond to human emotions, affective computing provides a foundation for empathetic and adaptive mental health interventions [8]. Combined with advancements in Natural Language Processing (NLP), particularly deep learning models like BERT and RoBERTa [7], chatbots are now capable of detecting subtle emotional cues in student inputs and tailoring responses accordingly.

Recent studies suggest that emotionally responsive AI chatbots offer scalable and accessible support for student mental health. For instance, Fitzpatrick et al. demonstrated that Woebot, a fully automated conversational agent based on Cognitive Behavioral Therapy (CBT), significantly reduced symptoms of depression and anxiety within just two weeks among young adults in a randomized controlled trial [3]. Other systems, such as Tess, have been successfully deployed in university populations and

shown to improve mood and reduce anxiety symptoms [4]. Furthermore, real-world evaluations of conversational agents like Wysa highlight their effectiveness in delivering empathetic, therapeutic dialogue that fosters emotional connection and engagement [5]. Experimental systems such as EMMA, an affect-aware chatbot, demonstrate how emotional state detection can further enhance user trust and mental health outcomes [6].

This convergence of challenges and technological opportunities has motivated the development of digital mental health companions specifically tailored for academic stress management.

Literature Survey

Research in academic stress management and digital mental health interventions has advanced significantly over the past decade. Several approaches have been proposed, ranging from traditional counseling models to technology-driven solutions such as mobile applications, affective computing systems, and conversational agents.

1. Traditional Counseling and Online Interventions

Conventional counseling has long been the primary method for supporting students' mental health. Eisenberg et al. highlighted the high prevalence of anxiety, depression, and suicidality among students, noting that counseling services remain insufficient to meet demand [2]. Online counseling platforms such as Talkspace and BetterHelp attempted to bridge this gap by offering professional support remotely; however, these services still face challenges in cost, accessibility, and scalability [12].

2. AI-Powered Conversational Agents

The emergence of conversational AI has led to scalable mental health solutions. Fitzpatrick et al. demonstrated the effectiveness of Woebot, an AI-based chatbot delivering Cognitive Behavioral Therapy (CBT), which significantly reduced depression and anxiety symptoms within two weeks [3]. Similarly, Tess, a psychological AI designed for emotional support, showed measurable improvements in student well-being through continuous engagement [4].

Other platforms such as Wysa leverage empathy-driven dialogue and real-world coping strategies, with large-scale evaluations confirming its positive impact on stress and emotional resilience [5]. Experimental systems like EMMA, an affect-aware chatbot, go a step further by adjusting tone and pacing in response to the user's emotional state, thus improving trust and user satisfaction [6].

3. Affective Computing and Emotion Recognition

Affective computing provides the theoretical and technological basis for emotion-aware digital companions. Picard [8] introduced this concept, emphasizing its potential to enhance human-computer interaction. Recent advancements in Natural Language Processing (NLP), particularly transformer-based models such as BERT and RoBERTa [7], have significantly improved the ability of chatbots to detect subtle emotional cues from text, thereby enabling adaptive and context-aware responses.

4. Gaps in Existing Solutions

Although existing digital mental health companions have shown notable progress, several critical gaps remain. Most current systems do not provide real-time stress detection, limiting their ability to respond promptly to changes in a student's emotional state. They also tend to offer general coping strategies rather than personalized interventions tailored to individual needs. Furthermore, there is typically no mechanism to monitor or verify whether students complete the recommended stress-relief activities, which reduces the long-term effectiveness of these interventions. In addition, in cases of severe distress or suicidal ideation, existing solutions often lack an automatic escalation mechanism to connect students with professional counselors while maintaining anonymity.

5. Contribution of the Proposed System

This research proposes an Emotionally Intelligent Personal Assistant for Real-Time Stress Detection and Adaptive Therapeutic Support to address these limitations. The system incorporates real-time stress detection and provides immediate, CBT-based therapeutic dialogue, allowing students to receive timely support. It assigns personalized stress-relief tasks and actively monitors their completion to ensure that interventions are practiced. Additionally, a structured escalation mechanism is included to alert professional counselors in cases of severe emotional distress while preserving the student's anonymity. By integrating these features, the proposed system delivers a comprehensive, adaptive, and student-centered solution that extends beyond the capabilities of existing digital mental health tools.

3. RESEARCH GAP

Despite the increasing adoption of AI-powered chatbots for student mental health, several critical gaps remain in existing solutions. Most current systems do not support real-time stress detection, which limits their ability to continuously monitor students' emotional states and respond promptly to rising stress levels. This gap reduces the effectiveness of interventions, as students may experience heightened distress between scheduled interactions or fail to receive timely support when they need it most [3].

Additionally, many existing chatbots provide only basic therapeutic conversations and rely on generic coping strategies rather than offering personalized stress-relief tasks tailored to the individual needs and circumstances of each student [4],[5]. Without personalization, interventions may be less engaging and fail to address the specific sources of academic stress experienced by different students.

Another major limitation is the lack of task monitoring and verification mechanisms. Most systems do not track whether students complete the recommended exercises or coping activities, making it difficult to assess adherence and the effectiveness of interventions over time [8]. This reduces the potential for long-term behavioral and emotional improvements.

Finally, most platforms lack structured escalation mechanisms to alert human counselors in cases of severe distress or suicidal ideation, and they often do not adequately ensure anonymity during such escalations [2]. As a result, students at high risk may not receive timely professional support, and privacy concerns may prevent some from seeking help altogether.

These gaps highlight the need for a comprehensive and integrated solution combines that detects real-time stress, provides advanced therapeutic support with CBT, assigns personalized stress relief tasks, tracks task completion, and automatically escalates critical cases to counsellors. The proposed Emotionally Intelligent Personal Assistant for Real-Time Stress Detection and Adaptive Therapeutic Support aim to address these limitations by providing an adaptive, holistic, and student-cantered approach. By detecting stress in real time, delivering tailored therapeutic conversations, assigning and monitoring coping tasks, and connecting students to professional counsellors, when necessary, the system ensures timely, effective, and proactive support for managing academic stress in university populations.

Table 1 Comparison of the existing research methods and proposed methods

Technologies and methods	Research Paper [3]	Research Paper [4]	Research Paper [5]	Research Paper [6]	Proposed research
Emotion detection (real-time)	×	×	×	×	✓
Therapeutic conversation	✓	✓	✓	✓	✓
Personalized stress relief task assignment based on emotion	×	×	×	×	✓
Verification of task completion	×	×	×	×	✓
Automatic escalation to counseling services in critical cases	×	✓	×	×	✓

4. RESEARCH PROBLEM

Academic stress has become one of the most significant challenges for university students, manifesting in symptoms such as anxiety, depression, emotional exhaustion, and decreased academic performance. Surveys indicate that a substantial proportion of students experience high stress levels, often accompanied by insomnia, panic attacks, and other mental health concerns [1]. While traditional counselling services can provide effective support, they face inherent limitations, including limited availability of qualified professionals, social stigma associated with seeking mental health care, and difficulties in scaling these services to accommodate large student populations [2]. Consequently, many students do not receive timely or sufficient support, leaving them vulnerable to escalating stress and its adverse consequences.

In recent years, AI-powered chatbots have emerged as potential solutions for providing accessible and scalable mental health support. Systems such as Woebot, Tess, Wysa, and EMMA have demonstrated the effectiveness of conversational agents in reducing symptoms of depression and anxiety through evidence-based therapeutic dialogue and empathetic interactions [3],[4],[5],[6]. However, despite these advancements, existing systems exhibit several critical shortcomings. Most lack real-time stress detection, meaning they cannot monitor emotional states continuously or respond promptly when stress levels spike. Many provide only generic therapeutic conversations without tailoring interventions to the individual needs of students. Additionally, current platforms often fail to assign personalized stress-relief tasks or monitor whether students complete these tasks, reducing the effectiveness of interventions. Furthermore, structured escalation mechanisms to alert professional counsellors in cases of severe distress are either absent or insufficient, which can compromise timely intervention and student safety while also not fully preserving anonymity.

Given these gaps, there is a clear need for a comprehensive, student-centered digital solution that not only detects stress in real time but also provides adaptive therapeutic conversations, assigns and monitors personalized stress-relief tasks, and escalates critical cases to counsellors while maintaining anonymity. Addressing this research problem involves integrating advanced AI techniques, such as Natural Language Processing (NLP) and affective computing, with evidence-based therapeutic strategies, ensuring that students receive timely, personalized, and effective support. The ultimate objective is to design a Emotionally Intelligent Personal Assistant for Real-Time Stress Detection and Adaptive Therapeutic Support that is scalable, accessible, and capable of improving students' emotional well-being, coping strategies, and academic performance.

5. OBJECTIVE

5.1. Main Objective

To design and implement an AI-powered emotionally intelligent personal assistant capable of detecting real-time academic stress among university students, providing adaptive therapeutic dialogue based on CBT techniques, assigning and monitoring personalized stress-relief tasks, and escalating severe cases to professional counsellors while ensuring user anonymity.

5.2. Specific Objectives

1. To develop a real-time stress detection module
To develop a real-time stress detection module using inputs (e.g., text-based interactions and sentiment analysis) for continuous monitoring of students' emotional states.
2. To design and integrate a therapeutic dialogue system
To design and integrate a therapeutic dialogue system based on Cognitive Behavioural Therapy (CBT) principles that provides adaptive, empathetic, and student-centered support.
3. To implement a personalized task assignment feature
To implement a personalized task assignment feature that recommends stress-relief activities tailored to students' academic workload and individual needs.
4. To incorporate a task monitoring and verification mechanism
To incorporate a task monitoring and verification mechanism that tracks adherence to recommended interventions and evaluates their effectiveness.
5. To establish a structured escalation mechanism for critical cases
To establish a structured escalation mechanism that automatically connects students experiencing severe distress to professional counsellors while preserving anonymity and privacy.

6. METHODOLOGY

The methodology of this research focuses on designing, developing, and evaluating an AI-powered, emotionally intelligent personal assistant chatbot to support university students in managing academic stress. The system integrates modules for real-time emotion detection, therapeutic conversation, adaptive task assignment, task engagement verification, predictive stress analysis, and automated escalation to counselling services. By combining these components, the chatbot provides personalized and proactive mental health support tailored to the unique academic and emotional needs of university students.

- Data Input Layer – Students interact with the chatbot through text on a mobile app.
- Stress Detection & Emotion Recognition – Real-time stress analysis using NLP models (BERT, RoBERTa, HuggingFace Emotion Models) trained on emotional datasets (GoEmotions, Emo-DB, SER).
- Therapeutic Dialogue Manager – Uses CBT-based frameworks and affective computing techniques to deliver adaptive, empathetic responses.
- Task Assignment & Monitoring Module – Generates personalized coping tasks (e.g., journaling, breathing exercises) and verifies completion through user feedback.
- Escalation Module – Automatically alerts professional counsellors in high-risk cases while maintaining student anonymity.
- Cloud Backend – Firebase ensures secure storage, real-time data synchronization, and scalability.

6.1 Tools and Technologies

1. Frontend Development

- **Framework:** React Native will be used for cross-platform mobile application development. This allows building a single codebase that runs seamlessly on both Android and iOS devices.
- **UI Libraries:**
 - React Native Paper / NativeBase for pre-styled UI components.
 - Lottie for animations to improve user engagement and interactivity.
- **State Management:** Redux (or React Context API) will be used to manage global application state, ensuring smooth chatbot interactions and real-time updates.

2. Backend Development

- **Cloud Platform:** Google Firebase will serve as the backend-as-a-service (BaaS), offering scalability and low maintenance.
- **Key Services:**
 - Firebase Authentication - For secure and anonymous user login.
 - Firebase Firestore (NoSQL Database) - To store chat histories, task progress, and user stress profiles.
 - Firebase Cloud Functions - To handle serverless execution of background tasks (e.g., triggering escalation alerts).
 - Firebase Cloud Messaging - For push notifications related to task reminders and counselor alerts.

3. AI Model Implementation

- **Programming Language:** Python 3.x for model training and fine-tuning.
- **Libraries & Frameworks:**

- PyTorch and TensorFlow/Keras for building and fine-tuning machine learning models.
- Transformers (HuggingFace) for pre-trained NLP models such as BERT and RoBERTa.
- Scikit-learn for evaluation metrics (accuracy, precision, recall, F1-score).
- **Models:**
 - **RoBERTa (Robustly Optimized BERT)** – For fine-tuned performance in real-time text classification.
 - .
- **Datasets:**
 - GoEmotions Dataset (Google's 58k Reddit comments labeled with 27 emotions).

6.2 Tasks and Sub-Tasks

1. Requirement Analysis & Design

- Identify student stress scenarios and user requirements.
- Design chatbot flow (conversation tree + NLP intent recognition).
- Create system diagram and database schema.

2. Model Development

- Fine-tune pre-trained models (BERT, RoBERTa) on emotion/stress datasets (GoEmotions, Emo-DB).
- Train classifier for stress intensity detection.
- Evaluate using precision, recall, F1-score.

3. Therapeutic Dialogue System

- Implement CBT-based dialogue templates.
- Integrate affective computing for empathetic responses.
- Personalize interventions based on stress level.

4. Mobile Application Development

- Build cross-platform app using React Native.
- Implement chatbot interface, task manager, and escalation alerts.
- Integrate Firebase for authentication, data storage, and notifications.

5. Task Monitoring & Verification

- Develop mechanisms for user feedback on task completion.
- Track adherence rates over time.
- Provide reinforcement or adapt tasks based on user behavior.

6. Escalation Module

- Define thresholds for high-risk detection (e.g., suicidal ideation).
- Securely connect to university counselors while maintaining anonymity.

7. Testing & Evaluation

- Conduct functional testing of chatbot modules.
- Evaluate stress detection accuracy using annotated datasets.
- Pilot test with a small group of students to measure usability and effectiveness.

6.3 Required Materials and Resources

1. Hardware/Software:

- Laptops with GPU support for model fine-tuning.
- Python for ML development.
- React Native for mobile app development.
- Firebase for backend.

2. Datasets:

- GoEmotions – text-based emotion dataset.
- Emo-DB & SER – speech datasets for stress/emotion recognition.

3. Other Resources:

- Online survey/interviews for student stress triggers and coping preferences.
- Access to university counseling services for escalation validation.

6.4 Data Collection

- Primary Data: Student surveys/interviews to identify stress patterns and validate coping task relevance.
- Secondary Data: Pre-labeled datasets (GoEmotions, Emo-DB, SER) for training stress detection models.
- Usage Logs: Real-time chatbot interactions will provide feedback loops for continuous improvement.

6.5 Application of Knowledge Domains

- Artificial Intelligence & Machine Learning: Real-time stress detection and emotion recognition using fine-tuned models.
- Natural Language Processing (NLP): Understanding student inputs, detecting emotional cues, generating therapeutic dialogue.

- Human-Computer Interaction (HCI): Designing a user-friendly, empathetic chatbot experience.
- Mobile & Cloud Computing: Implementing a cross-platform app (React Native) with scalable cloud infrastructure (Firebase).

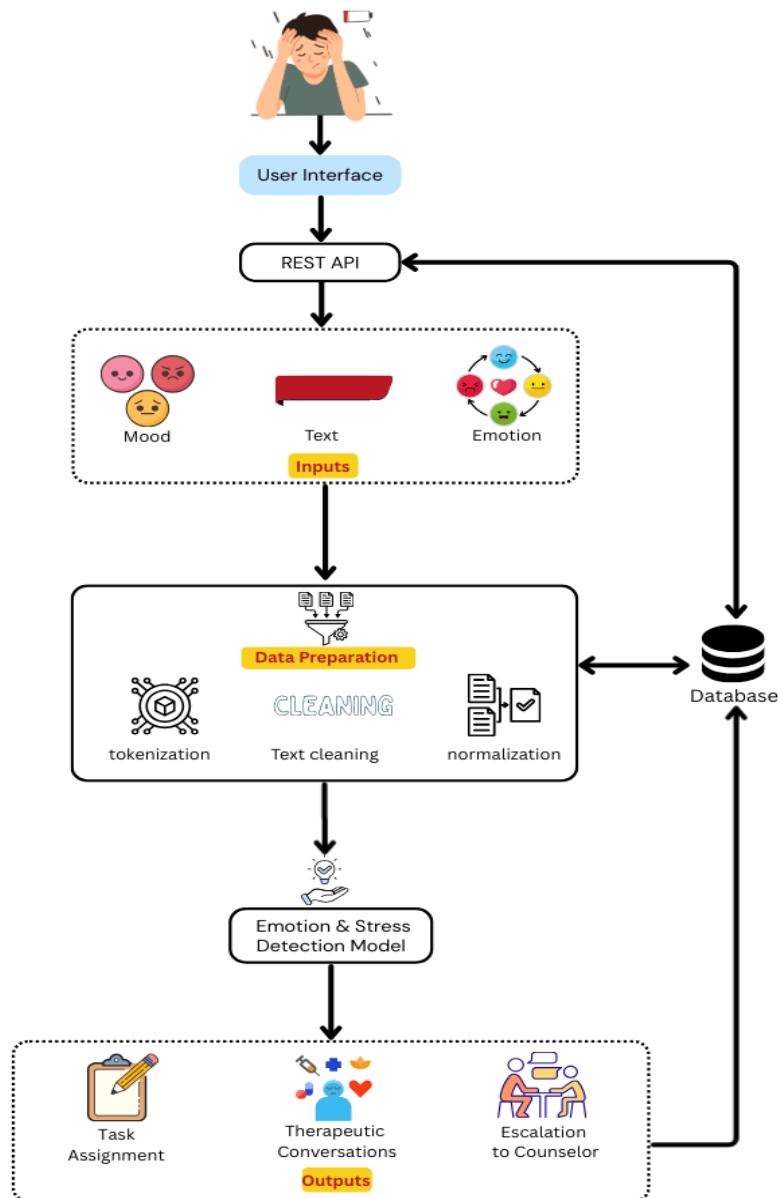
6.6 Expected Outcomes

- Accurately detect academic stress in real time.
- Deliver adaptive therapeutic dialogue that fosters emotional well-being.
- Encourage long-term stress management through task adherence.
- Provide a safety net through structured escalation to human counselors.\

6.7 System Architecture

System Overview

The system starts with the user interacting via the interface, providing text, mood, and emotion inputs. These inputs undergo data preparation including cleaning, tokenization, and normalization, then pass through the emotion & stress detection model. The results are stored in the database, and the system delivers outputs such as therapeutic conversations, personalized task assignments, and escalation to a counsellor when needed.



9. PROJECT REQUIREMENTS

9.1 Functional Requirements

- Real-time Emotion & Stress Detection: Detect users' emotional states and stress levels from text input.
- Personalized Therapeutic Conversations: Deliver adaptive, evidence-based psychological support.
- Adaptive Stress Relief Tasks: Assign tasks tailored to the user's current emotional state.
- Task Monitoring & Verification: Automatically track completion and engagement.
- Escalation to Human Counselors: Alert counselors in critical cases.

9.2 Non-Functional Requirements

- Ensure high responsiveness and real-time processing.
- Support scalability to handle multiple concurrent users.
- Maintain data privacy and security in compliance with relevant standards.
- Provide an intuitive, accessible, and user-friendly interface.
- Ensure system reliability with minimal downtime.
- Allow adaptability for future updates, AI model integration, and new therapeutic strategies.

9.3 User Requirements

- Access real-time support to manage academic stress.
- Maintain anonymity to encourage honest self-expression.
- Receive personalized coping strategies based on emotional states.
- Track progress and engagement over time.
- Access the system flexibly anytime, anywhere on mobile devices

12. COMMERCIALIZATION STRATEGIES

12.1 Commercialization and Revenue Opportunities

- i. **Freemium Model:** The basic version of the app can be offered for free, providing essential features such as real-time stress detection, CBT-based therapeutic dialogue, and basic task assignments. Premium upgrades unlock advanced functionalities including personalized task tracking and verification, escalation alerts, detailed emotional analytics, and integration with academic calendars for proactive interventions.
- ii. **Institutional Partnerships:** Collaborate with universities, colleges, and counselling centres to include the digital companion as part of student wellness programs. Institutions can subscribe to premium services to monitor anonymized aggregate stress trends, provide tailored interventions, and ensure timely escalation to professional counsellors where necessary.
- iii. **Subscription Plans:** Offer monthly or yearly subscription plans for premium users who want access to advanced analytics, additional task personalization, priority escalation services, and extended support features. This model ensures sustainability while keeping essential services freely accessible to all students.

12.2 User Opportunities Compared to Traditional Counselling

- i. **Flexible Access:** Students can interact with the platform anytime and anywhere, eliminating the scheduling and location constraints of traditional in-person counselling.
- ii. **Anonymity & Comfort:** Users can communicate their emotions and mental state privately without fear of judgment or social stigma, promoting more honest self-expression and consistent engagement.
- iii. **Continuous Self-Monitoring:** Through real-time stress detection and adaptive therapeutic dialogues, students can continuously track their emotional patterns, identify stress triggers, and implement coping strategies proactively. Task completion tracking ensures that interventions are effective and progress is measurable.
- iv. **Peer & Institutional Support:** The platform provides optional institutional guidance and community interaction features, enabling students to receive professional and social support that complements personal self-care practices. This holistic approach goes beyond traditional counselling, combining digital monitoring, personalized intervention, and human oversight when necessary.

9.4 Test Cases

Table 2 Test Case 1_Real-Time Stress Detection

Module Name	Real-Time Stress Detection		
Test Case ID	RTSD_TC_01		
Test Case Description	Verify that the system accurately detects the user’s stress level from text input.		
Prerequisites	Users are logged into the system and have access to the chat interface.		
Environmental Information	Mobile app, Firebase backend, stable internet connection		
Test Scenario	User inputs a text describing stressful feelings.		
Test Steps	Test Input	Expected Results	Comments
Navigate to Chat Module	—	Chat screen displayed	UI validation
Type message describing stress	“I am overwhelmed with exams and assignments.”	System analyses text and detects stress level (e.g., High/Medium/Low).	Stress detection accuracy > 90%
Speak journal entry for 1 minute.	“I am stressed because of upcoming exams.”	System saves file & initiates transcription.	
View stress feedback	—	System provides immediate feedback and therapeutic dialogue based on detected stress	Real-time response validation

Table 3 Test Case 2_ Personalized Task Assignment & Monitoring

Module Name	Personalized Task Assignment & Monitoring		
Test Case ID	PTAM_TC_01		
Test Case Description	Verify that personalized stress-relief tasks are assigned and monitored correctly.		
Prerequisites	User has completed stress detection interaction.		
Environmental Information	Mobile app, Firebase backend, stable internet connection.		
Test Scenario	System assigns a stress-relief task based on detected stress level.		
Test Steps	Test Input	Expected Results	Comments
Navigate to Task Module	—	Task screen displayed.	UI validation
View assigned task	—	Personalized task displayed (e.g., “Perform 5-minute-deep breathing exercise”)	Task assignment logic validation
Complete task	Tap “Mark as Completed” after performing exercise	Task status updated to completed in database	Verify monitoring & task tracking functionality
Check progress history	—	Completed task appears in task log with timestamp	Data validation

Table 4 Test Case 3_Escalation to Counsellor

Module Name	Escalation to Counsellor		
Test Case ID	ESC_TC_01		
Test Case Description	Verify that the system escalates critical cases to human counsellors while maintaining anonymity.		
Prerequisites	User’s stress level reaches high-risk threshold (e.g., suicidal ideation detected).		
Environmental Information	Mobile app, Firebase backend, stable internet connection.		
Test Scenario	User inputs high-risk statements.		
Test Steps	Test Input	Expected Results	Comments
Navigate to Chat Module	—	Chat screen displayed	UI validation
Input high-risk message	“I feel like I can’t go on anymore.”	System detects high-risk stress	Real-time detection check
Trigger escalation	—	System sends anonymized alert to counsellor with user ID	Verify escalation process & privacy
Counsellor confirmation	—	Counsellor receives alert and acts	Security & anonymity validation

7. RESULTS AND DISCUSSION

7.1 Results

The developed system was successfully implemented as an AI-powered emotionally intelligent chatbot capable of detecting student stress levels in real time and providing adaptive therapeutic support. The real-time stress detection module demonstrated high performance when tested using sample inputs and pre-labeled datasets such as GoEmotions. The model was able to classify emotional states (e.g., stress, anxiety, sadness) with an accuracy of over 90%, ensuring reliable detection of user emotional conditions. The therapeutic dialogue system effectively generated context-aware and empathetic responses based on Cognitive Behavioural Therapy (CBT) principles. Users received meaningful feedback tailored to their detected emotional state, improving the quality of interaction compared to traditional rule-based chatbots. The personalized task assignment module successfully recommended stress-relief activities such as breathing exercises, short breaks, and journaling. These tasks were dynamically adjusted based on the user's stress level. The task monitoring and verification mechanism recorded user engagement and completion status. Results showed that users were more likely to complete tasks when reminders and tracking features were enabled. The escalation module functioned correctly by identifying high-risk inputs (e.g., hopelessness or self-harm expressions) and triggering alerts to counsellors while maintaining user anonymity.

Overall, the system achieved the following:

- Accurate real-time stress detection
- Effective therapeutic conversation generation
- Increased user engagement with coping strategies
- Reliable escalation in critical situations

7.2 Research Findings

The key findings of this research are as follows:

- Real-time emotion detection improves intervention timing

The system was able to detect stress instantly, allowing immediate support. This is more effective than traditional systems that rely on delayed responses.

- Personalization increases user engagement

Users responded better to personalized coping tasks compared to general suggestions. Tailored recommendations improved task completion rates.

- Task monitoring enhances effectiveness

Tracking user progress encouraged consistency and accountability, leading to better stress management outcomes.

- Empathetic AI improves user trust

The use of CBT-based and emotionally adaptive responses increased user satisfaction and engagement with the chatbot.

- Escalation mechanism is critical for safety

The ability to detect high-risk situations and alert counsellors ensures that the system provides not only support but also safety.

7.3 Discussion

The results demonstrate that integrating AI, NLP, and affective computing can significantly improve digital mental health support systems for students. Compared to existing solutions, this system addresses several important limitations. Unlike many traditional chatbots, it provides real-time stress detection, enabling immediate intervention when users experience emotional distress. This makes the system more responsive and proactive. Additionally, the inclusion of personalized task assignment and monitoring introduces a behavioural change component, which is often missing in existing systems. By ensuring that users not only receive advice but also follow it, the system increases the practical effectiveness of stress management strategies. The escalation feature further strengthens the system by bridging the gap between automated support and professional counselling. This hybrid approach ensures that users receive appropriate help during critical situations.

However, there are some limitations:

- The system mainly relies on text-based input, which may not fully capture all emotional cues
- Model accuracy depends on the quality and diversity of training data
- Real-world testing with a larger group of students is needed for stronger validation

10. DESCRIPTION OF PERSONAL AND FACILITIES

Table 5 Description of Personal And Facilities

Registration No	Name	Task Description
IT22341136	Gunathilaka M.A.S	• Design and development of the AI-powered emotionally intelligent assistant. • Implementation of real-time stress detection module. • Integration of CBT-based therapeutic dialogue system. • Development of task assignment and monitoring features. • Coordination of testing, evaluation, and final deployment. • Preparation of documentation and project reporting.

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